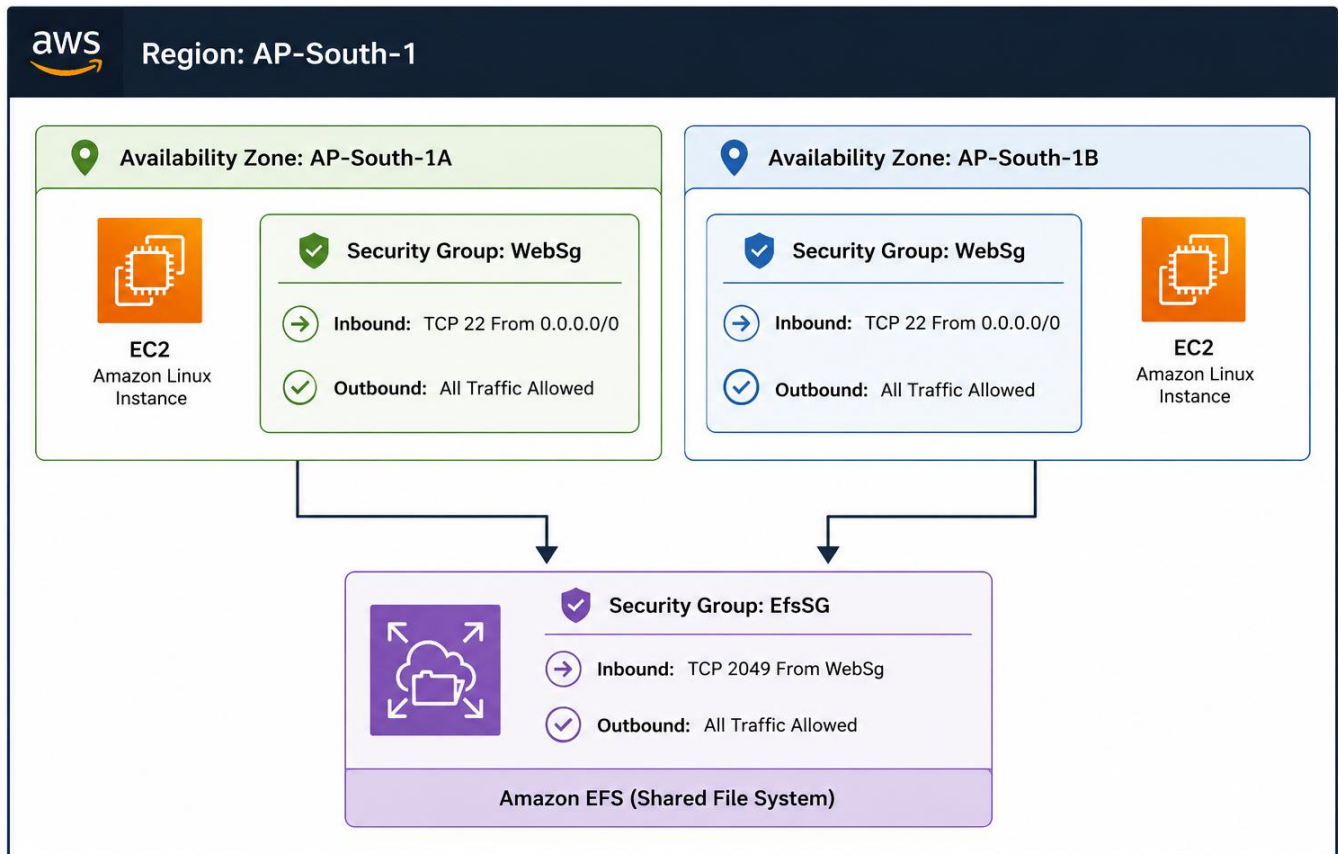


Elastic File Storage: -

Summary: -

Create Two Machine and learn how Elastic File Storage works as a Shared Storage.

Let's Create: -



- Let's Create 2 Security groups WebSg and EfsSG.
- For Creating Security groups open VPC go to Security group section and create security group.

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
SSH	TCP	22	A... 0.0.0.0/0	
Add rule				Delete

- Inbound rule of WebSg security group.

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
NFS	TCP	2049	Cu...	sg-0dd11073d84dd07d5	Delete

[Add rule](#)

- Inbound rule of EfsSG security group.

Security Groups (2) [Info](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

[WebSg](#) [EfsSG](#) [Clear filters](#)

☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-0dd11073d84dd07d5	WebSg	vpc-072c4acd761c3b942
☐	-	sg-02351d81fe5a9207c	EfsSG	vpc-072c4acd761c3b942

- Our Two Security group is ready.
- Now create two EC2 instance.

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-072c4acd761c3b942 (default) [Refresh](#)

Subnet [Info](#)

subnet-04ffa8e5e9c79ad3d [Refresh](#)

VPC: vpc-072c4acd761c3b942 Owner: 463646775279
 Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
 IP addresses available: 4090 CIDR: 172.31.32.0/20

[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups [Refresh](#)

WebSg sg-0dd11073d84dd07d5 [X](#)
 VPC: vpc-072c4acd761c3b942

[Compare security group rules](#)

- 1st EC2 OS Amazon Linux t3.micro
- Network Setting of 1st instance subnet – ap-south-1a, Security group – WebSg.

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-072c4acd761c3b942
172.31.0.0/16

(default) ▼



Subnet | [Info](#)

subnet-0bc545e3030c41ef4
VPC: vpc-072c4acd761c3b942 Owner: 463646775279
Availability Zone: ap-south-1b (aps1-az3) Zone type: Availability Zone
IP addresses available: 4090 CIDR: 172.31.0.0/20

↻ Create new subnet [↗](#)

Auto-assign public IP | [Info](#)

Enable ▼

Additional charges apply when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups | [Info](#)

Select security groups ▼

↻ Compare security group rules

WebSg sg-0dd11073d84dd07d5 ✕
VPC: vpc-072c4acd761c3b942

- 2nd EC2 OS Amazon Linux t3.micro
- Network Setting of 2nd instance Subnet – ap-south-1b, Security group – WebSg

Instances (2) [Info](#)

Last updated
less than a minute ago



Connect

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set ▼

🔍 Find Instance by attribute or tag (case-sensitive)

< 1 >



☐	Name 🔗 ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availabili
☐	EFS_VM2	i-06c49021e829b9ab0	Running 🔍 🔍	t3.micro	3/3 checks passed	ap-south-
☐	EFS_VM1	i-06fad2c2d1b0062c8	Running 🔍 🔍	t3.micro	3/3 checks passed	ap-south-

- Both instances are created.
- Now we will create Elastic File System (EFS).
- Open Aws console in new tab and search EFS and open it and click Create EFS.

Create file system

×

Create a file system with the recommended settings shown below by choosing Create file system. To view all settings or to customize your file system, choose Customize. [Learn more](#)

Name - optional
Name your file system.

Shared_Storage

Name can include letters, numbers, and +-=._:/ symbols, up to 256 characters.

Virtual Private Cloud (VPC)
Choose the VPC where you want EC2 instances to connect to your file system.

vpc-072c4acd761c3b942
default

Recommended settings

Your file system is created with the following recommended settings unless you choose to customize the file system. You will be charged for storage and throughput. We recommend reviewing pricing for these features using the [AWS Pricing Calculator](#).

Setting	Value	Editable after creation
Throughput mode Learn more	Elastic	Yes

CancelCustomizeCreate file system

- Now Enter name and click Customize.

Automatic backups
Automatically backup your file system data with AWS Backup using recommended settings. Additional pricing applies. [Learn more](#)

☐ Enable automatic backups

 We recommend that you create a backup policy for your file system

- Off Automatic Backup.

Encryption
Choose to enable encryption of your file system's data at rest. Uses the AWS KMS service key (aws/elasticfilesystem) by default. [Learn more](#)

☐ Enable encryption of data at rest

- Off Encryption.

► Tags optional

CancelNext

- Simply click Next.

Mount targets

A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#)

Availability zone

ap-south-1a

Subnet ID

subnet-04ffa8e5e9c79ad3d

IP address type

IPv4 only

IPv4 address

Optional

IPv6 address

-

Security groups

Choose security groups

sg-02351d81fe5a9207c
EfsSG

Remove

- Add 2 Mount targets because we have two ec2 instance in two different Availability Zone (ap-south-1a and ap-south-1b).
- For both add security group EfsSG and remove previous security group that added default.

Availability zone

ap-south-1b

Subnet ID

subnet-0bc545e3030c41ef4

IP address type

IPv4 only

IPv4 address

Optional

IPv6 address

-

Security groups

Choose security groups

sg-02351d81fe5a9207c
EfsSG

Remove

Add mount target

Cancel

Previous

Next

- Then scroll down click Next.
- Then again in next step simply click Next.
- Then in next step scroll down and click Create.

Success!
 File system (fs-0b39533f7f07de0a7) is available.

View file system

File systems (1)

View details

Delete

Create file system

<

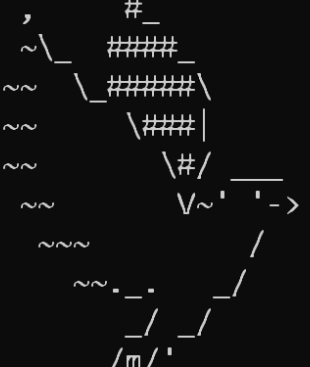
1

>

	Name ▾	File system ID ▾	Encrypted ▾	Total size ▾	Size in Standard ▾	Size in IA ▾	Size Archived ▾
☐	Shared Storage	fs-0b39533f7f07de0a7	☒ Unencrypted	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes

- Now EFS is created.
- Now Open CMD in your system.

```
C:\ ec2-user@ip-192-168-0-45:~  
Microsoft Windows [Version 10.0.19045.7184]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Akash Kumar>cd Downloads  
  
C:\Users\Akash Kumar\Downloads>ssh -i demo.pem ec2-user@13.207.2.165  
The authenticity of host '13.207.2.165 (13.207.2.165)' can't be established.  
ED25519 key fingerprint is SHA256:u5ZyuJ65FhjTP9UYG5J45trh5J/NX2FQilFNogIIE4s.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '13.207.2.165' (ED25519) to the list of known hosts.
```



```
,      #_  
~\_   #####_        Amazon Linux 2023  
~~ \#####\  
~~ \|###|  
~~ \|#/_____ https://aws.amazon.com/linux/amazon-linux-2023  
~~ V~'->  
~~~~  
~-.-./--/  
_/m/'
```

```
[ec2-user@ip-192-168-0-45 ~]$
```

- Now My Key that I created at EC2 creation is available in downloads folder so firstly I go inside it.
- Then `ssh -i {key.pem} ec2-user@{public-ip-address}`.
- Now authenticity is approved.


```
root@ip-172-31-35-27:~
=====Install 2 Packages

Total download size: 6.9 M
Installed size: 25 M
Is this ok [y/N]: yes
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm          4.3 MB/s | 156 kB    00:00
(2/2): amazon-efs-utils-3.1.1-1.amzn2023.x86_64.rpm    58 MB/s | 6.7 MB    00:00
-----Total
MB/s | 6.9 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
  Running scriptlet: stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
  Installing     : amazon-efs-utils-3.1.1-1.amzn2023.x86_64 2/2
  Running scriptlet: amazon-efs-utils-3.1.1-1.amzn2023.x86_64 2/2
  Verifying      : amazon-efs-utils-3.1.1-1.amzn2023.x86_64 1/2
  Verifying      : stunnel-5.58-1.amzn2023.0.2.x86_64 2/2

Installed:
  amazon-efs-utils-3.1.1-1.amzn2023.x86_64          stunnel-5.58-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-35-27 ~]# mkdir efs_
```

- Now Enter for create file {mkdir efs}
- Now go to you Efs tab in your browser and open created EFS.

Shared_Storage (fs-0b39533f7f07de0a7)

[Delete](#)[Attach](#)

General

[Edit](#)

Amazon resource name (ARN)

 arn:aws:elasticfilesystem:ap-south-1:463646775279:file-system/fs-0b39533f7f07de0a7

Performance mode

General Purpose

Throughput mode

Elastic

Lifecycle management

Transition into Infrequent Access (IA):

30 day(s) since last access

Transition into Archive: 90 day(s) since last access

Transition into Standard: None

Automatic backups

 Disabled

Encrypted

No

File system state

 Available

DNS name

 fs-0b39533f7f07de0a7.efs.ap-south-1.amazonaws.com

Replication overwrite protection

 Enabled

- Now click Attach.

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☒ Mount via DNS

☐ Mount via IP

Using the EFS mount helper:

```
sudo mount -t efs -o tls fs-0b39533f7f07de0a7:/ efs
```

Using the NFS client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-0b39533f7f07de0a7.efs.ap-south-1.amazonaws.com:/ efs
```

- Now Copy EFS Mount Helper.

```
root@ip-172-31-35-27:~#
=====Install 2 Packages
Total download size: 6.9 M
Installed size: 25 M
Is this ok [y/N]: yes
Downloading Packages:
(1/2): stunnel-5.58-1.amzn2023.0.2.x86_64.rpm          4.3 MB/s | 156 kB    00:00
(2/2): amazon-efs-utils-3.1.1-1.amzn2023.x86_64.rpm    58 MB/s | 6.7 MB    00:00
-----Total
MB/s | 6.9 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
  Running scriptlet: stunnel-5.58-1.amzn2023.0.2.x86_64 1/2
  Installing     : amazon-efs-utils-3.1.1-1.amzn2023.x86_64 2/2
  Running scriptlet: amazon-efs-utils-3.1.1-1.amzn2023.x86_64 2/2
  Verifying      : amazon-efs-utils-3.1.1-1.amzn2023.x86_64 1/2
  Verifying      : stunnel-5.58-1.amzn2023.0.2.x86_64 2/2

Installed:
  amazon-efs-utils-3.1.1-1.amzn2023.x86_64          stunnel-5.58-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-35-27 ~]# mkdir efs
[root@ip-172-31-35-27 ~]# sudo mount -t efs -o tls fs-0b39533f7f07de0a7:/ efs_
```

- Now paste it in cmd.
- Now it will mount my EFS with my EC2 instances.
- And this shared storage is accessible from my EC2 instance.
- How I can verify.

```
root@ip-172-31-35-27:~/efs
[root@ip-172-31-35-27 ~]# cd efs
[root@ip-172-31-35-27 efs]# ls
[root@ip-172-31-35-27 efs]#
```

- Now enter `cd efs` go inside efs folder that we created.
- Check what present inside with `ls`.
- Nothing is present.

```
root@ip-172-31-35-27:~/efs
[root@ip-172-31-35-27 ~]# cd efs
[root@ip-172-31-35-27 efs]# ls
[root@ip-172-31-35-27 efs]# cat >> new
This File is created for VM1
^Z
[1]+  Stopped                  cat >> new
[root@ip-172-31-35-27 efs]#
```

- Now We create a file by { `cat << new` }.
- Then enter some thing and press `ctrl + Z`

```
root@ip-172-31-35-27:~/efs
[root@ip-172-31-35-27 efs]# ls
new
[root@ip-172-31-35-27 efs]#
```

- Now enter ls.
- Now you see a new folder is available that we created now.

```
root@ip-172-31-35-27:~/efs
[root@ip-172-31-35-27 efs]# ls
new
[root@ip-172-31-35-27 efs]# cat new
This File is created for VM1
[root@ip-172-31-35-27 efs]#
```

- Open another cmd and repeat previous code for 2nd EC2 instance.
- And create a file inside it cat >> VM2 and access it from 1st instance CMD.

```
root@ip-172-31-35-27:~/efs
[root@ip-172-31-35-27 efs]# cat VM2
This is a VM2
[root@ip-172-31-35-27 efs]#
```

- Now you see its Working.